

Time: All in Your Mind

Clocks measure equal seconds. Human beings do not experience them equally. Research shows how mood, attention, memory and hope can reshape the felt movement of a life.

BY MAYA CHELLE

On its clock face, time appears to be an objective features of existence. Divided it into equal seconds, organized into neat boxes on the calendar. Yet, measured time does not equate experienced time. An enjoyable afternoon goes by in a moment, while a the last hour at the office feels interminable.

A universal feeling. Also a scientifically tested one.

Psychologists distinguish between estimating a duration and the feeling of the passage of time. The first means determining whether an interval lasted ten seconds or twenty. The second means asking whether those seconds felt fast, slow, lively, empty or suspended. This difference is central to understanding why emotional states change time, without ever changing the clock.

Research on depression demonstrates this effect. Current literature on depression has suggested that depressed individuals lose their sense of time. These findings reveal a nuance: depression alters the feeling of passage of time without altering the objective estimation process. Past and future play vital roles in the brain's morphed sense of the present.

Let's dive deeper.

Two kinds of time

Laboratory studies typically examine interval timing through tasks such as estimation. A participant might be asked to reproduce a tone's duration or decide which of two intervals was longer. These tasks test chronometric performance.

Passage-of-time judgments ask how quickly time seems to move. Note the distinction. People with depression often show performance on objective timing tasks broadly similar to that of non-depressed participants. However they also report that time drags. This measure is far more subjective and emotionally-bound.

A meta-analysis by Thönes and Oberfeld combined 16 studies involving 433 participants with depression and 485 controls. Depression had a medium-sized effect on the subjective flow of time: participants with depression tended to experience time as passing more slowly. Yet the analysis found no significant overall effect on the four principal laboratory timing tasks (Thönes & Oberfeld, 2015).

16

Studies combined in the
time-perception meta-analysis

918

Participants across depressed and
control groups

$g = 0.66$

Effect on the subjective flow of time

The result suggests that a person can know how long an interval lasted while suffering through how that interval felt. Subjective time stems from differences in by attention, emotion, activity, expectation and meaning.

Lived time

A 2018 study led by David Vogel investigated these experiences in depth. The researchers recruited 25 patients receiving inpatient treatment at the University Hospital Cologne. Participants had been diagnosed with a severe depressive episode and screened for non-chronic major depressive disorder (Vogel et al., 2018).

Instead of restricting participants to numerical scales, the researchers used an open-ended Time Questionnaire. It asked about the passage of time and about the meanings of the past, present and future. Participants could write as much or as little as they considered necessary.

The responses produced 235 statements, which the researchers organized into 30 categories through qualitative content analysis. Independent coding produced strong agreement, with a Krippendorff's alpha of .93 (Vogel et al., 2018). The study therefore focused both on how quickly time seemed to pass and on how time was lived from inside a life.

WHAT 'LIVED TIME' MEANS

Lived time concerns whether life feels as if it is progressing, standing still, repeating itself or moving beyond one's control. It includes the relationship among memory, present action and anticipated possibility.

A dead end

Fifteen of the 25 participants reported a general slowing of time. Nine described its passage as inexorable, endless, circular or at a standstill. Five said that time had lost meaning or could no longer be properly felt, while six reported that pleasant or social activities could improve or normalize its passage (Vogel et al., 2018).

The present itself was unstable. Ten participants described it as extended, while another ten experienced it as a point rather than a duration. Nine connected it with negative emotions, and five described it as meaningless. Seven believed they could influence the present; seven believed it was outside their control (Vogel et al., 2018).

These apparently contradictory answers show that depression does not produce one uniform clock. It can make the present feel both endless and strangely empty, or so compressed that it no longer functions as a bridge between past and future.

Ordinarily, the present receives some of its meaning from what it can change. People work now because the work may produce a later result. When the future no longer feels reachable, present action loses part of its direction. Chronological time continues, but psychological movement weakens.

The problem may not be that the present no longer seems to lead anywhere.

Attention changes the clock

A study by Sévigny, Everett and Grondin compared 15 participants classified as depressed with 20 non-depressed participants. The depressed group made more omissions on a sustained-attention task. They also performed more poorly when discriminating relatively long intervals of 1,120 and 1,280 milliseconds, while differences did not appear for shorter pairs of intervals. Their production of one- and ten-second intervals was also more variable (Sévigny et al., 2003).

The authors argued the role of attentional demands in processing longer intervals. This offers one possible mechanism: timing depends partly on the cognitive resources available to monitor and organize an interval.

Gil and Droit-Volet used a temporal bisection task in which participants classified signals lasting between 400 and 1,600 milliseconds as short or long. Higher depression scores were associated with judging the signals as shorter, although

temporal sensitivity was similar across groups. Measures of sadness helped explain part of the pattern (Gil & Droit-Volet, 2009).

These experimental results show that 'time perception' is not a single ability. Attention, interval length, mood and task design can produce different outcomes.

Daily life

A 2024 experience-sampling study followed 105 participants in daily life. Five times per day, participants evaluated how time had passed during the previous 30 minutes. At 11 p.m., they evaluated the day as a whole (Liu et al., 2024).

For the shorter 30-minute judgments, alertness, time expectation, emotional valence and arousal all predicted how quickly time seemed to pass. For judgments of the entire day, time expectation and emotional arousal remained predictive, while the other factors were less stable. The overall judgment of a day was most closely related to both the average and the most recent within-day reports (Liu et al., 2024).

The study highlights two principles. First, subjective time depends on more than whether a mood is positive or negative. Alertness and expectations play a role. Second, retrospective judgments are constructed from an accumulation of moments, but they are not viewed objectively. They may be disproportionately influenced by what happened recently.

WHY IT MATTERS

Time can feel slow when it violates our expectation of change. Waiting becomes especially difficult when we monitor time closely but cannot act to produce the event we want. Action feels productive. Waiting can be miserable.

Waiting

Rankin, Sweeny and Xu studied undergraduate students waiting for examination grades and law graduates waiting for bar-exam results. Across three studies, people who experienced time as moving slowly tended to report more worry, anxiety, negative emotion and difficulty coping. The evidence also tentatively supported the reverse path: greater distress predicted a slower experience of time (Rankin et al., 2019).

This suggests a possible feedback loop. Distress makes time feel slower; slow time leaves more room to monitor uncertainty; that monitoring may intensify distress. The researchers described the association as potentially bidirectional. (Rankin et al., 2019).

The finding shows that the emotional shape of time changes during ordinary periods of uncertainty. What matters is not only how long people wait, but whether they possess information, control and meaningful activity while waiting.

The past

Depression is also associated with changes in autobiographical memory. Overgeneral autobiographical memory occurs when a person asked to recall one specific event instead produces a category of repeated experiences or an extended period. For example, rather than remembering one particular disappointing conversation, the person may recall that conversations 'always go badly.'

A 2023 meta-analysis by Weiss-Cowie, Verhaeghen and Duarte synthesized 67 published and unpublished works. Compared with controls, people with depression showed lower autobiographical-memory specificity, with Hedges' $g = -0.73$, and greater categoricity, with $g = 0.77$. The pattern was strongest in current clinical major depression but remained detectable in subthreshold and remitted groups (Weiss-Cowie et al., 2023).

Overgeneral memory concerns the level of detail and the ability to retrieve a bounded episode. Specific memories can support problem-solving because they provide concrete information about what

happened, what was tried and what changed. A general conclusion such as 'nothing ever works' offers fewer details from which to construct a new response.

67 Works synthesized in the memory meta-analysis	$g = -0.73$ Reduced autobiographical specificity	$g = 0.77$ Increased categorical remembering
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A bleak outlook

Remembering the past and imagining the future rely on related constructive processes. To picture a possible future, the mind recombines elements of previous experience. If specific details are difficult to retrieve, positive future scenes may also become less vivid or plausible.

Hallford and colleagues compared 117 people experiencing a major depressive episode with 47 participants without one. Participants imagined personally relevant positive future events. Those in the depressed group generated events with less specificity, detail, vividness, mental imagery and first-person perspective. They also rated those events as less plausible and reported less anticipatory pleasure (Hallford et al., 2020).

Within the depressed group, lower detail and vividness, weaker mental imagery and lower personal significance each uniquely predicted reduced pleasure while anticipating the event. The finding does not prove that imagery deficits cause anhedonia, but it shows that hope is partly sensory and constructive: a future that cannot be pictured clearly may be difficult to desire (Hallford et al., 2020).

This helps distinguish having no future from having a future that feels psychologically unavailable. A person may understand that positive events are possible in the abstract while struggling to simulate one as concrete, personal and likely.

Hope is not only a belief that something good could happen. It is also the ability to imagine that event as vivid, personal and reachable.

A blocked future

In Vogel and colleagues' study, ten participants associated the future with fear, worry or hopelessness. Eight described it as diminished, blocked, meaningless or no longer part of subjective experience. Seven characterized it as uncertain, impossible to plan or beyond their control (Vogel et al., 2018).

A blocked future does not necessarily mean that a person believes time will end. It means that future possibilities no longer exert their usual emotional pull. Advice such as 'look forward' or 'make a plan' assumes that the future is visible as a field of meaningful action. That assumption may not hold during a depressive episode.

The findings were not absolute. Ten participants also associated the future with hope. Depression can narrow possibility without erasing every positive expectation (Vogel et al., 2018).

The gaps

These studies differ substantially. Some recruited patients with diagnosed major depression; others measured depressive symptoms in non-clinical samples. Some examined milliseconds in a laboratory; others analyzed written descriptions, memories or imagined futures. Results vary, and no single universal claim can be derived. However, tendencies are noted.

The Vogel study was small, qualitative and restricted to patients with severe depression who were receiving heterogeneous treatments. Its method was not designed to establish causation or represent every form of depression

(Vogel et al., 2018). The Thönes and Oberfeld meta-analysis noted that medication and dopamine-related mechanisms had not been investigated consistently enough to evaluate fully (Thönes & Oberfeld, 2015).

The future-thinking study did not assess cognitive functioning, and the comparison group's history of previous depressive episodes was unknown (Hallford et al., 2020). The daily-life study involved a non-clinical sample and described general influences on passage-of-time judgments rather than depression specifically (Liu et al., 2024).

These limitations do not invalidate the evidence. Instead, they define what can responsibly be concluded: mood is associated with meaningful changes in subjective time, autobiographical memory and future simulation, but no single experience is universal and no one task captures the entire phenomenon.

A CLINICAL CAUTION

Research findings describe group-level tendencies. They cannot diagnose an individual, explain every episode of depression or replace assessment by a qualified professional.

So, subjective time?

Subjective time affects more than comfort. If the present feels endless and the future inaccessible, making plans is a struggle. If the past is remembered mainly in broad negative categories, it may offer fewer concrete exceptions and fewer usable solutions. If positive futures lack detail and plausibility, effort toward them may feel unrewarding.

Behaviors that look from the outside like passivity, laziness, or a lack of motivation may therefore occur within a different temporal world. Passivity may mean the future does not exert its usual pull and the present does not feel capable of changeable.

The research also reveals something more general about perception. Human beings interpret time.

Measured time remains indispensable. It coordinates societies and allows experiments to compare behavior. But lived time explains why two people inhabit the same hour differently. One may experience an interval as temporary and meaningful; another may experience it as closed, repetitive and without an exit.

Understanding this difference encourages empathy. To consider the world from another's mind. Through not only their eyes, but their sense of time.

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